To: <Recipient's_Email_Address>

From: DMD <medintel@hcpconnectsX.com>

Subject line 1: Is it asthma or lung NET? Read the case study inside

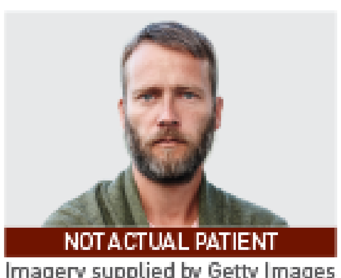
Subject line 2: Could their nonspecific respiratory symptoms be lung NET? A case study

Subject line 3: Lung NET mimics common respiratory conditions—read a case study

Subject line 4: Treatment with AFINITOR® (everolimus) Tablets: a case study

Case study: Delayed lung NET diagnosis

Learn more about treatment with AFINITOR® (everolimus) Tablets for patients with progressive, well-differentiated, nonfunctional lung NET with unresectable, locally advanced, or metastatic disease.



Imagery supplied by Getty Images

Most lung NET patients present with advanced disease due to delays in diagnosis[1,2]

Samuel is a 45-year-old white male with no history of smoking

2 years ago
Experienced coughing and wheezing consistent with asthma

18 months ago
Initially diagnosed with asthma

16 months ago
Prescribed a course of glucocorticosteroids

12 months ago
Prescribed a short-acting β_2-agonist when symptoms persisted

9 months ago
Samuel had an episode of hemoptysis[1]

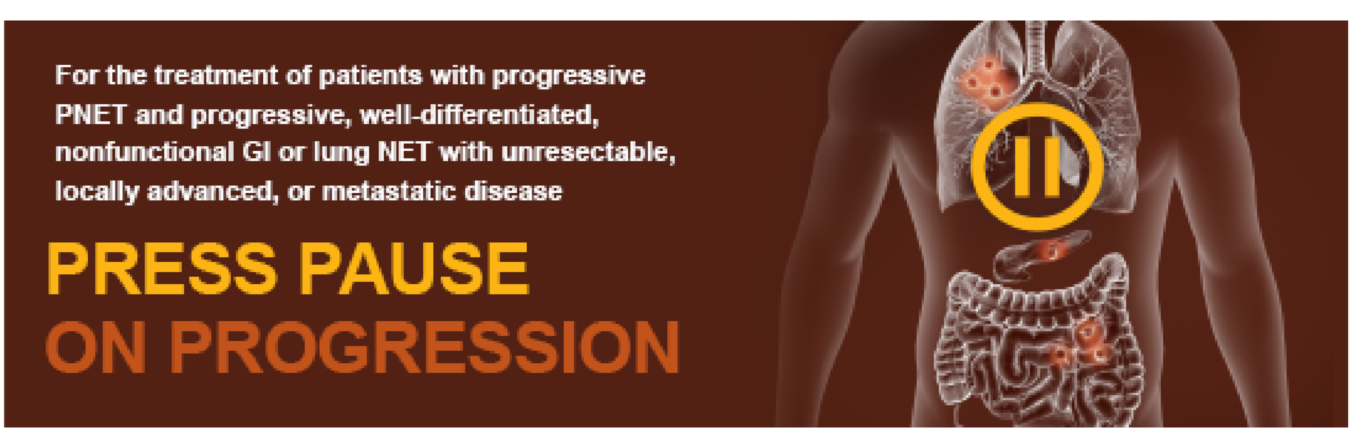
6 months ago
A chest CT revealed a mass in the central lung[3]

3 months ago
IHC staining from fine-needle biopsy confirmed grade 1 typical carcinoid (TC)[3]

Samuel was diagnosed with advanced, well-differentiated lung NET and prescribed AFINITOR by his oncologist

Abbreviations: CT, computed tomography; IHC, immunohistochemical; NET, neuroendocrine tumors.

If you see a patient with a history of unexplained respiratory symptoms, consider progressive, nonfunctional lung NET[1]



Median PFS was 11.0 months (95% CI, 8.4-13.9) for AFINITOR vs 4.6 months (95% CI, 3.1-5.4) for placebo (HR=0.35 [95% CI, 0.27-0.45], *P*<0.001) in PNET.[4]

Median PFS was 11.0 months (95% CI, 9.2-13.3) for AFINITOR vs 3.9 months (95% CI, 3.6-7.4) for placebo (HR=0.48 [95% CI, 0.35-0.67], *P*<0.001) in GI and lung NET.[4]

Abbreviations: GI, gastrointestinal; PFS, progression-free survival; PNET, neuroendocrine tumors of pancreatic origin.

INDICATIONS

AFINITOR is indicated for the treatment of adults with progressive neuroendocrine tumors of pancreatic origin (PNET) with unresectable, locally advanced, or metastatic disease.

AFINITOR is indicated for the treatment of adults with progressive, well-differentiated, nonfunctional neuroendocrine tumors (NET) of gastrointestinal (GI) or lung origin with unresectable, locally advanced, or metastatic disease.

AFINITOR is not indicated for the treatment of patients with functional carcinoid tumors.

IMPORTANT SAFETY INFORMATION

AFINITOR is contraindicated in patients with hypersensitivity to everolimus, to other rapamycin derivatives, or to any of the excipients.

Noninfectious Pneumonitis: Noninfectious pneumonitis was reported in up to 19% of patients treated with AFINITOR; some cases reported with pulmonary hypertension (including pulmonary arterial hypertension) as a secondary event. The incidence of Common Terminology Criteria (CTC) grade 3 and 4 noninfectious pneumonitis was up to 4.0% and up to 0.2%, respectively. Fatal outcomes have been observed. Monitor for clinical symptoms or radiological changes. Opportunistic infections such as *Pneumocystis jiroveci* pneumonia (PJP) should be considered in the differential diagnosis. Manage noninfectious pneumonitis by dose interruption until symptoms resolve, follow with a dose reduction, and consider the use of corticosteroids. Discontinue AFINITOR if toxicity recurs at grade 3 or for grade 4 cases. For patients who require use of corticosteroids, prophylaxis for PJP may be considered. The development of pneumonitis has been reported even at a reduced dose.

Infections: AFINITOR has immunosuppressive properties and may predispose patients to bacterial, fungal, viral, or protozoal infections (including those with opportunistic pathogens). Localized and systemic infections, including pneumonia, mycobacterial infections, other bacterial infections; invasive fungal infections such as aspergillosis, candidiasis, or PJP; and viral infections, including reactivation of hepatitis B virus, have occurred. Some of these infections have been severe (eg, leading to sepsis, respiratory failure, or hepatic failure) or fatal. Physicians and patients should be aware of the increased risk of infection with AFINITOR. Treatment of preexisting invasive fungal infections should be completed prior to starting treatment with AFINITOR. Be vigilant for signs and symptoms of infection and institute appropriate treatment promptly; interruption or discontinuation of AFINITOR should be considered. Discontinue AFINITOR if invasive systemic fungal infection is diagnosed and institute appropriate antifungal treatment.

PJP has been reported in patients who received everolimus, sometimes with a fatal outcome. This may be associated with concomitant use of corticosteroids or other immunosuppressive agents; consider prophylaxis for PJP when concomitant use of these agents is required.

Angioedema With Concomitant Use of Angiotensin-Converting Enzyme (ACE) Inhibitors: Patients taking concomitant ACE inhibitor therapy may be at increased risk for angioedema (eg, swelling of the airways or tongue, with or without respiratory impairment). In a pooled analysis, the incidence of angioedema in patients taking everolimus with an ACE inhibitor was 6.8% compared to 1.3% in the control arm with an ACE inhibitor.

Oral Ulceration: Mouth ulcers, stomatitis, and oral mucositis have occurred in patients treated with AFINITOR at an incidence ranging from 44% to 78% across the clinical trial experience. Grade 3/4 stomatitis was reported in 4% to 9% of patients. In such cases, topical treatments are recommended, but alcohol-, hydrogen peroxide-, iodine-, or thyme-containing mouthwashes should be avoided. Antifungal agents should not be used unless fungal infection has been diagnosed.

Renal Failure: Cases of renal failure (including acute renal failure), some with a fatal outcome, have been observed in patients treated with AFINITOR.

Impaired Wound Healing: Everolimus delays wound healing and increases the occurrence of wound-related complications like wound dehiscence, wound infection, incisional hernia, lymphocele, and seroma. These wound-related complications may require surgical intervention. Exercise caution with the use of AFINITOR in the perisurgical period.

Laboratory Tests and Monitoring: Elevations of serum creatinine and proteinuria have been reported. Renal function (including measurement of blood urea nitrogen, urinary protein, or serum creatinine) should be evaluated prior to treatment and periodically thereafter, particularly in patients who have additional risk factors that may further impair renal function.

Hyperglycemia, hyperlipidemia, and hypertriglyceridemia have been reported. Blood glucose and lipids should be evaluated prior to treatment and periodically thereafter. More frequent monitoring is recommended when AFINITOR is coadministered with other drugs that may induce hyperglycemia. Management with appropriate medical therapy is recommended. When possible, optimal glucose and lipid control should be achieved before starting a patient on AFINITOR.

Reductions in hemoglobin, lymphocytes, neutrophils, and platelets have been reported. Monitoring of complete blood count is recommended prior to treatment and periodically thereafter.

Drug-Drug Interactions: Avoid coadministration with strong CYP3A4/PgP inhibitors (eg, ketoconazole, itraconazole, clarithromycin, atazanavir, nefazodone, saquinavir, telithromycin, ritonavir, indinavir, nelfinavir, voriconazole). Use caution and reduce the AFINITOR dose to 2.5 mg daily if coadministration with a moderate CYP3A4/PgP inhibitor is required (eg, amprenavir, fosamprenavir, aprepitant, erythromycin, fluconazole, verapamil, diltiazem). Avoid coadministration with strong CYP3A4/PgP inducers (eg, phenytoin, carbamazepine, rifampin, rifabutin, rifapentine, phenobarbital); however, if coadministration is required, consider doubling the daily dose of AFINITOR using increments of 5 mg or less.

Hepatic Impairment: Exposure to everolimus was increased in patients with hepatic impairment. For patients with severe hepatic impairment (Child-Pugh class C), AFINITOR may be used at a reduced dose if the desired benefit outweighs the risk. For patients with mild (Child-Pugh class A) or moderate (Child-Pugh class B) hepatic impairment, a dose reduction is recommended.

Vaccinations: The use of live vaccines and close contact with those who have received live vaccines should be avoided during treatment with AFINITOR.

Embryo-Fetal Toxicity: Fetal harm can occur if AFINITOR is administered to a pregnant woman. Advise pregnant women of the potential risk to a fetus. Advise female patients of reproductive potential of the potential risk to a fetus and to use effective contraception while using AFINITOR and for 8 weeks after ending treatment.

Adverse Reactions in Advanced, Progressive PNET: The most common adverse reactions (incidence ≥30%) were stomatitis (70%), rash (59%), diarrhea (50%), fatigue (45%), edema (39%), abdominal pain (36%), nausea (32%), fever (31%), headache (30%), and decreased appetite (30%). The most common grade 3/4 adverse reactions (incidence ≥5%) were stomatitis (7%) and diarrhea (5.5%). Deaths primarily due to adverse events during the double-blind treatment phase occurred in 7 patients taking AFINITOR.

Laboratory Abnormalities in Advanced, Progressive PNET: The most common laboratory abnormalities (incidence ≥50%, all grades) were: decreased hemoglobin (86%) and bicarbonate (56%); increased fasting glucose (75%), alkaline phosphatase (74%), cholesterol (66%), and aspartate transaminase (56%). The most common grade 3/4 laboratory abnormalities (incidence ≥5%) were: decreased hemoglobin (15%), lymphocytes (16%), and phosphate (10%), and increased glucose (17%) and alkaline phosphatase (8%).

Adverse Reactions in Advanced, Progressive, Well-Differentiated, Nonfunctional GI and Lung NET: The most common adverse reactions (incidence ≥30%) were stomatitis (63%), infections (58%), diarrhea (41%), peripheral edema (39%), fatigue (37%), and rash (30%). The most common grade 3/4 adverse reactions (incidence ≥5%) were infections (11%), stomatitis (9%), diarrhea (9%), fatigue (5%), and hyperglycemia (5%).

Laboratory Abnormalities in Advanced, Progressive, Well-Differentiated, Nonfunctional GI and Lung NET: The most common laboratory abnormalities (incidence ≥50%, all grades) were anemia (81%), hypercholesterolemia (71%), lymphopenia (66%), elevated aspartate transaminase (57%), and hyperglycemia (55%). The most common grade 3/4 laboratory abnormalities (incidence ≥5%) were lymphopenia (17%), hyperglycemia (6%), elevated alanine transaminase (6%), hypokalemia (6%), and anemia (5%).

Click here for full Prescribing Information.

References: 1. Wolin EM. Challenges in the diagnosis and management of well-differentiated neuroendocrine tumors of the lung (typical and atypical carcinoid): current status and future considerations. *Oncologist.* 2015;20(10):1123-1131. **2.** Rekhtman N. Neuroendocrine tumors of the lung: an update. *Arch Pathol Lab Med.* 2010;134(11):1628-1638. **3.** Kunz PL, Reidy-Lagunes D, Anthony LB, et al. Consensus guidelines for the management and treatment of neuroendocrine tumors. *Pancreas.* 2013;42(4):557-577. **4.** Afinitor [prescribing information]. East Hanover, NJ: Novartis Pharmaceuticals Corp; 2016.

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If you see a patient with a history of unexplained respiratory symptoms, consider progressive, nonfunctional lung NET[1]

For the treatment of patients with progressive PNET and progressive, well-differentiated, nonfunctional GI or lung NET with unresectable, locally advanced, or metastatic disease

PRESS PAUSE ON PROGRESSION

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